

OPMIG-06: Processing, Parsing & Transformation

Series: OPMIG | **Notebook:** 6 of 9 | **Created:** December 2025

OpenPipeline Migration Series | Notebook 6 of 9

Level: Intermediate to Advanced

Estimated Time: 75 minutes

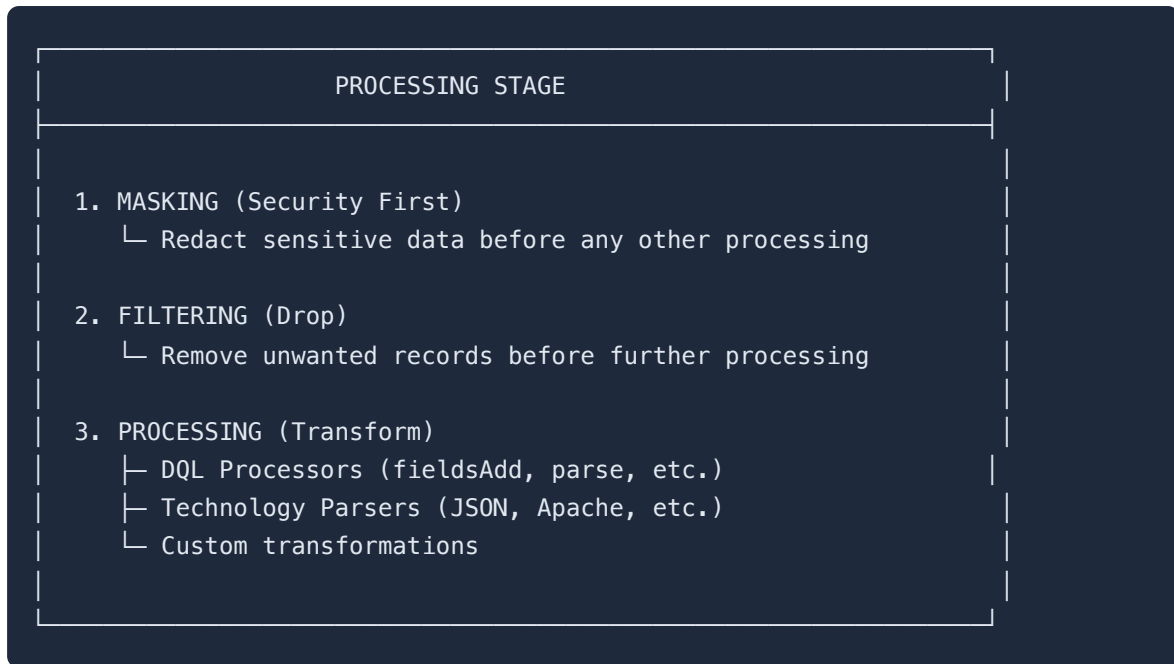
Learning Objectives

By completing this notebook, you will:

1. Master Dynatrace Pattern Language (DPL) for log parsing
 2. Configure DQL processors for data transformation
 3. ★ **NEW:** Parse Apache, Nginx, and JSON logs with production-ready patterns
 4. ★ **NEW:** Migrate from ELK/Logstash (Grok to DPL conversion)
 5. ★ **NEW:** Use the complete Parsing Pattern Library (timestamps, stack traces, HTTP)
 6. Implement drop processors for cost optimization
 7. Validate parsing success rates and troubleshoot failures
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Processing Stage Overview

The Processing stage is where data transformation happens. It includes three sub-stages executed in order:



Processor Execution Order

Within each sub-stage, processors execute in the order they're defined. You can reorder them in the UI.

💡 **Best Practice:** Order processors logically - parse first, then enrich with computed fields based on parsed values.

DQL Processor Commands

The DQL processor supports a subset of DQL commands for data transformation.

Available Commands

Command	Purpose	Example
<code>fieldsAdd</code>	Add new fields	<code>fieldsAdd env = "production"</code>
<code>fieldsRemove</code>	Remove fields	<code>fieldsRemove sensitive_field</code>
<code>fieldsRename</code>	Rename fields	<code>fieldsRename old = new_name</code>
<code>parse</code>	Extract with DPL	<code>parse content, "INT:count"</code>

fieldsAdd Examples

Static Values

```
fieldsAdd environment = "production"
fieldsAdd application = "checkout-service"
fieldsAdd team = "platform-engineering"
```

Conditional Values (if/else)

```
fieldsAdd severity = if(loglevel == "ERROR", "critical",
                        else: if(loglevel == "WARN", "warning",
                                else: "info"))
```

Computed Values

```
fieldsAdd message_length = stringLength(content)
fieldsAdd short_host = substring(host.name, 0, 15)
fieldsAdd is_error = loglevel == "ERROR"
```

String Operations

```
fieldsAdd normalized_status = toLowerCase(status)
fieldsAdd log_prefix = substring(content, 0, 50)
fieldsAdd clean_message = trim(content)
```

Coalesce (First Non-Null)

```
fieldsAdd effective_level = coalesce(loglevel, status, "UNKNOWN")
```

fieldsRemove Examples

```
// Remove single field
fieldsRemove internal_id

// Remove multiple fields
fieldsRemove temp_field, debug_info, internal_state
```

fieldsRename Examples

```
// Standardize field names
fieldsRename user_id = userId
fieldsRename request_id = requestId
fieldsRename transaction_id = transactionId
```

Dynatrace Pattern Language (DPL)

DPL is a powerful pattern matching language for extracting structured data from text.

Core Matchers

Matcher	Description	Matches
INT	Integer	42 , -17 , 0
LONG	Long integer	1234567890123
DOUBLE	Decimal	3.14 , -0.5 , 1.0
IPADDR	IP address	192.168.1.1 , ::1
IPV4ADDR	IPv4 only	10.0.0.1
IPV6ADDR	IPv6 only	2001:db8::1
LD	Line data (to delimiter)	Any text until next match
DATA	Greedy match	Everything remaining
SPACE	Whitespace	Spaces, tabs
NSPACE	Non-whitespace	Word-like content
WORD	Word chars	hello , user123
JSON	JSON object	{"key": "value"}
EOL	End of line	Line terminator

Pattern Syntax Elements

Syntax	Meaning	Example
MATCHER:field	Extract to named field	INT:count
MATCHER	Match but don't extract	SPACE
MATCHER?	Optional match	(':' INT:port)?
'literal'	Match exact text	'error_code='
(a b)	Alternatives	('user='\ 'userId=')
MATCHER{n,m}	Quantifier	WORD{1,3}

Basic Parse Examples

```
// Extract user ID after "user="
parse content, "'user=' LD:user_id"

// Extract error code (integer)
parse content, "'error_code=' INT:error_code"

// Extract IP and port
parse content, "IPADDR:client_ip ':' INT:port"

// Extract JSON payload
parse content, "LD JSON:payload"
```

Real-World Log Format Examples NEW

Production-ready DPL patterns for the most common log formats.

Apache Access Logs (Common Log Format)

Sample: 192.168.1.100 - frank [12/Dec/2024:10:30:45 +0000] "GET /api/users HTTP/1.1" 200 1234

```
parse content, ""
IPADDR:client_ip SPACE '-' SPACE LD:user SPACE
'[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):timestamp ']' SPACE
'"' LD:method SPACE LD:request_path SPACE LD:protocol '"' SPACE
INT:status_code SPACE INT:response_bytes
""
```

Extracted: client_ip, user, timestamp, method, request_path, protocol, status_code, response_bytes

Nginx (with Response Time)

Sample: 192.168.1.50 - - [12/Dec/2024:10:30:45 +0000] "POST /api/checkout HTTP/1.1" 201 456 "-" "Mozilla/5.0" "0.342"

```

parse content, ""
  IPADDR:client_ip SPACE '-' SPACE '-' SPACE
  '[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):timestamp ']' SPACE
  ''' LD:method SPACE LD:request_path SPACE LD:protocol ''' SPACE
  INT:status_code SPACE INT:response_bytes SPACE
  ''' LD:referrer ''' SPACE ''' LD:user_agent ''' SPACE
  ''' DOUBLE:response_time_sec '''
  ""
  | fieldsAdd response_time_ms = toInt(response_time_sec * 1000)

```

JSON Application Logs

Sample: {"timestamp":"2024-12-

12T10:30:45.123Z","level":"ERROR","service":"payment-api","message":"Payment gateway timeout"}

Option 1: Use Technology Parser (Recommended)

- Add **Technology** processor → Select **JSON** parser
- All fields automatically flattened

Option 2: DQL Parsing

```

parse content, "JSON:log_data"
  | fieldsAdd service = log_data["service"]
  | fieldsAdd message = log_data["message"]
  | fieldsAdd level = log_data["level"]

```

Syslog (RFC 3164)

Sample: <34>Dec 12 10:30:45 webserver sshd[1234]: Failed password for admin from 192.168.1.100

```

parse content, ""
  '<'; INT:priority '>';
  TIMESTAMP('MMM dd HH:mm:ss'):timestamp SPACE
  LD:hostname SPACE LD:app_name '[' INT:pid ']:' SPACE
  DATA:message
  ""
  | fieldsAdd facility = toInt(priority / 8)
  | fieldsAdd severity = toInt(priority % 8)

```

Java Application Logs (Log4j)

Sample: 2024-12-12 10:30:45,123 [http-nio-8080-exec-5] ERROR
com.example.Service - Payment failed

```
parse content, ""
  TIMESTAMP('yyyy-MM-dd HH:mm:ss,SSS'):log_timestamp SPACE
  '[' LD:thread ']' SPACE
  LD:level SPACE LD:logger SPACE '-' SPACE
  DATA:message
  ""
```

Stack Trace Extraction:

```
parse content, "LD:exception_class ':' SPACE LD:exception_message EOL"
| parse content, "'at ' LD:error_location '(' LD:file ':' INT:line_number
'|"
```

Kubernetes / Container Logs

Sample: 2024-12-12T10:30:45.123456789Z stdout F {"level":"info","msg":"Request processed"}

```
parse content, ""
  TIMESTAMP('yyyy-MM-dd\'T\'HH:mm:ss.SSSSSSSSSXXX'):k8s_timestamp SPACE
  LD:stream SPACE LD:log_tag SPACE
  JSON:log_payload
  ""
  | fieldsAdd level = log_payload["level"]
  | fieldsAdd msg = log_payload["msg"]
```

Common Parsing Patterns

Apache/Nginx Access Logs

Sample log:


```
192.168.1.100 - - [12/Dec/2024:10:30:45 +0000] "GET /api/users HTTP/1.1" 200 1234
```

DPL Pattern:

```
parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '['  
LD:timestamp ']' SPACE '\"' LD:method SPACE LD:path SPACE LD:protocol '\"'  
SPACE INT:status_code SPACE INT:bytes"
```

Key-Value Logs

Sample log:

```
userId=12345, action=login, status=success, duration=150ms
```

DPL Patterns:

```
// Extract each key-value pair  
parse content, "'userId=' INT:user_id"  
parse content, "'action=' LD:action ','"  
parse content, "'status=' LD:status ','"  
parse content, "'duration=' INT:duration_ms 'ms'"
```

Flexible User ID Extraction

Sample logs with varying formats:

```
Processing request for user=john123  
User userId=john123 authenticated  
Request from user_id=john123 received
```

DPL Pattern (alternatives):

```
parse content, "('user='|'userId='|'user_id=') LD:user_id"
```

Timestamp Parsing

DPL with timestamp format:

```
parse content, "TIMESTAMP('yyyy-MM-dd HH:mm:ss'):log_timestamp"  
parse content, "TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):apache_time"
```

Optional Fields

Sample log:

```
Request to server:8080 completed  
Request to server completed
```

DPL Pattern (optional port):

```
parse content, "'Request to ' LD:server (':' INT:port)? ' completed'"
```

Stack Trace Extraction

DPL Pattern:

```
parse content, "LD:exception_class ':' LD:exception_message"
```

Data Transformation Examples

Example 1: Parse and Enrich Application Logs

Input:

```
{"content": "[2024-12-12T10:30:45] ERROR PaymentService - Payment failed for  
orderId=12345, amount=99.99"}
```

Processor 1: Parse log structure

```
parse content, "'[' TIMESTAMP('yyyy-MM-dd'T\''HH:mm:ss'):log_time ']' SPACE  
LD:level SPACE LD:service ' - ' DATA:message"
```

Processor 2: Extract payment details

```
parse content, "'orderId=' INT:order_id ','"  
| parse content, "'amount=' DOUBLE:amount"
```

Processor 3: Add computed fields

```
fieldsAdd severity = if(level == "ERROR", "critical", else: "normal")
| fieldsAdd application_tier = "payment"
```

Example 2: JSON Log Processing

Input:

```
{"content": "{\"level\":\"info\",\"msg\":\"Request processed\",\"duration_ms\":150}"}
```

Processor: Parse JSON and flatten

```
parse_content, "JSON:json_data"
```

Using a Technology Parser (JSON) is often easier for JSON logs.

Example 3: Multi-Format Log Normalization

Processor: Normalize different log level formats

```
fieldsAdd normalized_level = if(contains(content, "ERROR") OR
contains(content, "error"), "ERROR",
                                else: if(contains(content, "WARN") OR
contains(content, "warn"), "WARN",
                                else: if(contains(content, "INFO") OR
contains(content, "info"), "INFO",
                                else: if(contains(content, "DEBUG") OR
contains(content, "debug"), "DEBUG",
                                else: "UNKNOWN"))))
```

Parsing Pattern Library NEW

Timestamp Patterns (10+ Formats)

Format	Example	DPL
ISO 8601	2024-12-12T10:30:45Z	<code>TIMESTAMP('yyyy-MM-dd\ 'T\ 'HH:mm:ssXXX')</code>
ISO + MS	2024-12-12T10:30:45.123Z	<code>TIMESTAMP('yyyy-MM-dd\ 'T\ 'HH:mm:ss.SSSXXX')</code>
Apache	12/Dec/2024:10:30:45+0000	<code>TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z')</code>
Syslog	Dec 12 10:30:45	<code>TIMESTAMP('MMM dd HH:mm:ss')</code>
Java	2024-12-12 10:30:45,123	<code>TIMESTAMP('yyyy-MM-dd HH:mm:ss,SSS')</code>
MySQL	2024-12-12 10:30:45	<code>TIMESTAMP('yyyy-MM-dd HH:mm:ss')</code>
Unix Epoch	1702380645	<code>LONG:epoch → toTimestamp(epoch * 1000)</code>

HTTP Request Patterns

```
// Full request line
parse content, "" LD:method SPACE LD:path SPACE LD:protocol ""

// Separate path and query
parse content, "" LD:method SPACE LD:path ('?' LD:query)? SPACE LD:protocol ""

// RESTful API paths (/api/v1/users/12345)
parse content, ""/api/v' INT:api_version '/' LD:resource '/' INT:id"
```

Key-Value Patterns

```
// Simple: user=john count=42
parse content, "'user=' LD:user SPACE 'count=' INT:count"

// Quoted: user="John Doe" email="john@example.com"
parse content, "'user=' LD:user ' ' SPACE 'email=' LD:email ' '"

// Logfmt: level=info msg="OK" duration=150ms
parse content, "'level=' LD:level SPACE 'msg=' LD:msg ' ' SPACE 'duration='
INT:dur 'ms'"
```

Stack Trace Patterns

```
// Java exception
parse content, "LD:exception_class ':' SPACE LD:exception_message EOL"

// Stack trace line
parse content, "'at ' LD:class_method '(' LD:file ':' INT:line ')"

// Caused by
parse content, "'Caused by: ' LD:caused_by ':' SPACE LD:message"

// Python traceback
parse content, "'File ' LD:file '", line ' INT:line ', in ' LD:function"
```

PII Masking Patterns

```
// Credit cards (1234-5678-9012-3456 → ****-****-****-****)
fieldsAdd content = replaceAll(content, "\\b\\d{4}[\\s-]?\\d{4}[\\s-]?\\d{4}[\\s-]?\\d{4}\\b", "****-****-****-****")

// Partial masking (keep last 4)
fieldsAdd content = replaceAll(content, "\\b(\\d{4})[\\s-]?(\\d{4})[\\s-]?(\\d{4})[\\s-]?(\\d{4})\\b", "****-****-****-$4")

// Email addresses
fieldsAdd content = replaceAll(content, "\\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\\.?[A-Z|a-z]{2,}\\b", "***@***.***")

// SSN (123-45-6789 → ***-**-****)
fieldsAdd content = replaceAll(content, "\\b\\d{3}-\\d{2}-\\d{4}\\b", "***-**-****")
```

Network Patterns

```
// IP and port
parse content, "IPADDR:ip ':' INT:port"

// IPv4 only
parse content, "IPV4ADDR:ipv4"

// IPv6 only
parse content, "IPV6ADDR:ipv6"

// URL parsing
parse content, "LD:protocol '://' LD:hostname (':' INT:port)? LD:path ('?' LD:query)?"
```

ELK/Logstash Migration Patterns NEW

Grok vs. DPL: Key Differences

Grok (Logstash)	DPL (OpenPipeline)
<code>%{IP:client_ip}</code>	<code>IPADDR:client_ip</code>
<code>%{INT:count}</code>	<code>INT:count</code>
<code>%{WORD:user}</code>	<code>WORD:user</code>
<code>%{GREEDYDATA:msg}</code>	<code>DATA:msg</code>
<code>\\s+</code>	<code>SPACE</code>

Apache Log Migration

Grok: `%{COMBINEDAPACHELOG}`

DPL:

```
parse content, ""
  IPADDR:client_ip SPACE LD:ident SPACE LD:auth SPACE
  '[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):timestamp ']' SPACE
  ''' LD:method SPACE LD:request_path (SPACE LD:http_version)? ''' SPACE
  INT:status_code SPACE (INT:response_bytes | '-') SPACE
  ''' LD:referrer ''' SPACE ''' LD:user_agent '''
  ""
```

Logstash Filter → OpenPipeline Mapping

Logstash Filter	OpenPipeline
<code>grok { }</code>	DQL parse processor
<code>mutate { add_field }</code>	<code>fieldsAdd field = "value"</code>
<code>mutate { remove_field }</code>	<code>fieldsRemove field</code>
<code>mutate { rename }</code>	<code>fieldsRename old = new</code>
<code>drop { }</code>	Drop processor
<code>json { }</code>	JSON Technology Parser
<code>date { }</code>	TIMESTAMP in parse

Complete Migration Example

Logstash:

```
filter {
  if [level] == "DEBUG" { drop { } }
  grok { match => { "message" => "%{TIMESTAMP_ISO8601:ts} %{LOGLEVEL:level} %{GREEDYDATA:msg}" } }
  mutate { add_field => { "env" => "production" } }
}
```

OpenPipeline:

1. Drop processor: `loglevel == "DEBUG"`
2. DQL parse: `parse content, "TIMESTAMP('yyyy-MM-dd HH:mm:ss'):ts SPACE LD:level SPACE DATA:msg"`
3. DQL add: `fieldsAdd env = "production"`

Technology Bundle Parsers

OpenPipeline includes built-in parsers for common log formats.

Available Technology Parsers

Parser	Log Format	Extracted Fields
Apache	Apache access logs	client_ip, method, path, status, bytes
Nginx	Nginx access logs	Similar to Apache
JSON	JSON-formatted logs	All JSON fields flattened
Syslog	RFC 3164/5424 syslog	facility, severity, hostname, message
Log4j	Java Log4j format	level, logger, thread, message
AWS CloudWatch	AWS logs	AWS-specific fields

When to Use Technology Parsers

Use Technology Parser	Use Custom DQL/DPL
Standard log format	Custom/proprietary format
Quick setup needed	Specific field extraction
Common technology	Conditional parsing
All fields needed	Selective extraction

Configuring Technology Parsers

1. Open pipeline in OpenPipeline settings
 2. Go to **Processing** tab
 3. Click **+ Processor** → **Technology**
 4. Select parser type
 5. Configure matching condition
 6. Save
-

Drop Processors

Drop processors remove records from the pipeline before storage.

Common Drop Patterns

Use Case	Matching Condition
Debug logs	<code>loglevel == "DEBUG"</code>
Trace logs	<code>loglevel == "TRACE"</code>
Health checks	<code>contains(content, "health")</code>
Readiness probes	<code>contains(content, "/ready")</code>
Metrics endpoints	<code>contains(content, "/metrics")</code>
Heartbeats	<code>contains(content, "heartbeat")</code>
Specific source	<code>log.source == "noisy-service"</code>

Drop Processor Configuration

```
Processor Type: Drop
Name: Drop debug logs
Matching Condition: loglevel == "DEBUG" OR status == "DEBUG"
```

Combining Drop Conditions

```
// Drop all non-essential logs
loglevel == "DEBUG"
  OR loglevel == "TRACE"
  OR contains(content, "health")
  OR contains(content, "/metrics")
```

 **Important:** Dropped data is gone forever. Test drop conditions carefully before deploying.

Advanced Processing Patterns

Pattern 1: Fix Missing Timestamp

When logs have a custom timestamp format that's not recognized:

```
// Parse timestamp from content
parse content, "'[' TIMESTAMP('yyyy-MM-dd HH:mm:ss'):parsed_timestamp ']'"
```

Pattern 2: Fix Missing Log Level

When logs don't have a standard loglevel field:

```
// Extract level from content
fieldsAdd loglevel = if(contains(content, "[ERROR]") OR contains(content,
"ERROR:"), "ERROR",
    else: if(contains(content, "[WARN]") OR
contains(content, "WARN:"), "WARN",
    else: if(contains(content, "[INFO]") OR
contains(content, "INFO:"), "INFO",
    else: if(contains(content, "[DEBUG]") OR
contains(content, "DEBUG:"), "DEBUG",
    else: "NONE"))))
```

Pattern 3: Parse JSON from Within Text

When JSON is embedded in a log message:

```
// Extract JSON from text
parse content, "LD JSON:embedded_json LD"
```

Pattern 4: Compute Response Time Categories

```
fieldsAdd response_category = if(duration_ms < 100, "fast",
    else: if(duration_ms < 500, "normal",
    else: if(duration_ms < 1000, "slow",
    else: "very_slow"))
```

Pattern 5: Standardize Boolean Fields

```
fieldsAdd is_success = if(status == "success" OR status == "ok" OR status == "200", true, else: false)
```

Pattern 6: Extract Service Name from Path

```
// From /api/v1/users → users  
parse content, "'/api/v' INT '/' LD:service_name"
```

Validating Your Processing

After configuring processors, validate that parsing is working correctly.

```
// Check parsing success rate across all pipelines  
fetch logs, from: now() - 1h  
| summarize {  
    total = count(),  
    with_loglevel = countIf(isNotNull(loglevel)),  
    with_status = countIf(isNotNull(status)),  
    with_either = countIf(isNotNull(loglevel) OR isNotNull(status))  
}  
| fieldsAdd parsing_rate = round((toDouble(with_either) / toDouble(total)) * 100, decimals: 1)
```

```
// Parsing success rate by pipeline  
fetch logs, from: now() - 1h  
| filter isNotNull(dt.openpipeline.pipelines)  
| summarize {  
    total = count(),  
    parsed = countIf(isNotNull(loglevel))  
}, by: {dt.openpipeline.pipelines}  
| fieldsAdd parsing_rate = round((toDouble(parsed) / toDouble(total)) * 100, decimals: 1)  
| sort parsing_rate asc
```

```
// Sample logs that failed parsing (no loglevel extracted)
fetch logs, from: now() - 1h
| filter isNull(loglevel) AND isNull(status)
| fields timestamp, log.source, dt.openpipeline.pipelines, content
| limit 25
```

```
// Verify specific parsed fields exist
// Replace 'your_field' with fields your parsing should create
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {
    total = count(),
    with_user_id = countIf(isNotNull(user_id)),
    with_request_id = countIf(isNotNull(request_id))
}, by: {dt.openpipeline.pipelines}
```

```
// Sample successfully parsed logs to verify field extraction
fetch logs, from: now() - 1h
| filter isNotNull(loglevel)
| limit 20
```

```
// Verify drop processors are working
// If drops are configured, debug log count should be low
fetch logs, from: now() - 1h
| summarize {debug_count = countIf(loglevel == "DEBUG")}
| fieldsAdd message = if(debug_count == 0, "✅ Drop processor working - no
debug logs",
                        else: "⚠️ Debug logs still present - check drop
configuration")
```

```
// Check log level distribution after processing
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {loglevel}
| sort log_count desc
```

DPL Architect Tool

Dynatrace provides a **DPL Architect** tool for building and testing patterns:

Accessing DPL Architect

1. Navigate to **Settings → OpenPipeline**
2. When adding a parse processor, click **Open DPL Architect**
3. Or access via: `https://{your-environment}.apps.dynatrace.com/ui/apps/dynatrace.dpl.architect`

Using DPL Architect

1. Paste sample log content
2. Build pattern interactively
3. See extracted fields in real-time
4. Copy pattern to processor definition

 **Tip:** Always test patterns in DPL Architect before deploying to production pipelines.

Complete Processing Pipeline Example

Pipeline: `application-logs`

Processor 1: Drop Debug (Drop)

```
Matching: loglevel == "DEBUG" OR contains(content, "[DEBUG]")
```

Processor 2: Parse Application Log (DQL)

```
parse content, "'[' TIMESTAMP('yyyy-MM-dd HH:mm:ss'):log_ts ']' SPACE '[' LD:level ']' SPACE '[' LD:thread ']' SPACE LD:class ' - ' DATA:message"
```

Processor 3: Extract Request ID (DQL)

```
parse content, "'requestId=' LD:request_id"
```

Processor 4: Add Environment Tags (DQL)

```
fieldsAdd environment = "production"
| fieldsAdd application = "checkout-service"
| fieldsAdd team = "platform"
```

Processor 5: Compute Severity (DQL)

```
fieldsAdd severity = if(level == "ERROR", "P1",
                        else: if(level == "WARN", "P2",
                                else: "P3"))
```

Next Steps

Now that you can transform data, continue with:

Notebook	Focus Area
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- [OpenPipeline Processing](#)
- [Processing Examples](#)
- [Dynatrace Pattern Language](#)
- [DPL Architect Tool](#)
- [DQL Functions in OpenPipeline](#)

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